

Unveiling and Mitigating Untargeted Poisoning Attacks on Federated Knowledge Graph Embedding

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Abstract

Federated knowledge graph embedding (FKGE) leverages distributed knowledge graphs (KGs) to collaboratively learn latent representations while preserving data privacy. However, this decentralized paradigm exposes vulnerabilities to client-side attacks, particularly untargeted poisoning attacks, which remain underexplored in FKGE and risk severe performance degradation. This paper presents the first systematic investigation of untargeted poisoning attacks on FKGE, unveiling their feasibility at both data and embedding levels. We design three novel poisoning attacks, among which the embedding-level attacks exhibit superior attack efficacy. To mitigate such threats, we further propose a learning-based defense mechanism that leverages embedding density for anomaly detection to identify malicious clients. Our theoretical results substantiate the effectiveness of both the attacks and the defense, and clarify why embedding-level poisoning is especially potent. Extensive experiments demonstrate that the proposed attacks induce significant performance deterioration, with core metrics dropping by up to 93%. Conversely, LDM effectively counters these attacks and restores performance to pre-attack levels or even surpasses them when mitigating embedding-level attacks.

CCS Concepts

• Computing methodologies → Knowledge representation and reasoning; • Security and privacy;

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Keywords

Knowledge Graph Embedding, Federated Learning, Poisoning Attacks, Robustness

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1 Introduction

Knowledge graphs (KGs) represent real-world connections through graph structures of entities and relations [16–19], playing key roles in recommender systems [39, 48] and information retrieval [38, 49]. However, their symbolic nature complicates downstream applications on the Web, where systems must bridge heterogeneous sources. *Knowledge graph embedding* (KGE) [35] addresses this by mapping entities and relations into continuous vector spaces [33]. Nevertheless, real-world KGs are distributed and privacy constrained across websites, platforms, and APIs, preventing direct sharing and posing a core Web challenge of learning from decentralized knowledge. *Federated Learning* (FL) [22, 25, 45, 46] addresses this by enabling collaborative training with data localization and model sharing, preserving privacy while handling distributed data. Based on this, *Federated KGE* (FKGE) [5] aggregates local entity embeddings to construct global ones, **enhancing KGE models** with multi-source knowledge while maintaining privacy.

FL typically assumes that clients are honest and cooperative, but this is often untenable in real-world scenarios, especially in competitive environments. For example, dominant participants may disrupt others to gain advantages, compromising FL's collaborative nature. Among the various threats, poisoning attacks are the most

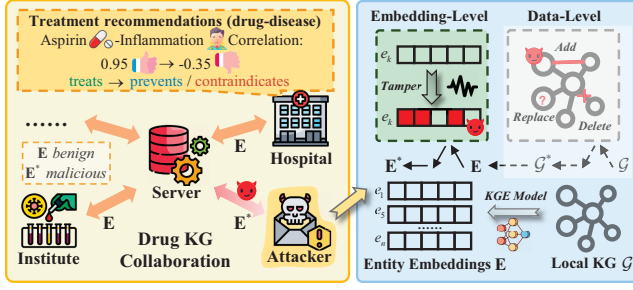


Figure 1: A case study of FL with drug KGs. Attackers influence other institutions by uploading malicious entity embeddings (e.g., the attacker may induce the misbelief that aspirin has preventive or contraindicated effects on inflammation). In FKGE, attackers can launch embedding-level attacks, by-passing the training process, which differs from data-level adversarial attacks in KGE research.

studied in FL [14, 21]. Current FKGE research has concentrated on targeted poisoning attacks [47], while **untargeted poisoning attacks** [10, 34, 41, 44] remain insufficiently explored. In multi-institutional drug KG collaborations (Fig. 1), attackers can upload polluted embeddings, degrading the inference accuracy of drug-disease relationships. Unlike targeted attacks on specific triples, untargeted attacks impair overall KGE performance, posing a significant real-world threat [44]. To this end, this paper investigates the following two critical questions: **I) How can diverse untargeted poisoning attacks on FKGE be designed from different levels? II) How can these attacks be effectively defended against?**

In response to Question I), from a technical viewpoint, clients train KGE models using local KGs and share knowledge by uploading entity embeddings. Unlike in standard KGE, attackers in FKGE could only influence others by uploading malicious embeddings, leading to diluted attack effects during aggregation. Besides tampering with local KGs at the **data level** (e.g., addition or deletion), attackers can also directly manipulate entity embeddings at the **embedding level** (see Fig. 1). Therefore, we conduct a comprehensive study of untargeted poisoning attacks on FKGE from both levels. Specifically, we propose a data poisoning attack based on similarity for entity replacement and two noise-based embedding poisoning attacks. Notably, embedding poisoning attacks are more direct and effective. This is because manipulating the embeddings can bypass the local training process and is agnostic to the type of KGE model.

Regarding Question II), we extend our investigation to defense against untargeted poisoning attacks on FKGE at both data and embedding levels. While KGE defenses typically focus on enhancing local training robustness [26, 43], these approaches are ineffective against embedding-level attacks, because such attacks are executed at post-training phase. In FL, robust aggregation is a common defense [2], but in FKGE, the structural heterogeneity of entity embeddings across clients makes direct analysis and traditional aggregation ineffective [5]. Therefore, it is necessary to develop a special defense mechanism for FKGE. Observing that the proposed attacks increase embedding sparsity, we design a **Learning-based Defense Mechanism (LDM)** that improves feature representation through **embedding density**. This method employs an anomaly

detection model trained on historical global embeddings to identify malicious clients, leveraging past training insights to reduce bias. The main contributions can be summarized as follows:

- **Problem.** We present the first holistic study for untargeted poisoning attacks, an underexplored yet critical aspect of FKGE research, encompassing both attack strategies and the corresponding defense mechanism.
- **Attack.** We propose three untargeted poisoning attacks to unveil the vulnerabilities of FKGE, including one data poisoning attack and two embedding poisoning attacks. To the best of our knowledge, this work represents the first attempt to design untargeted poisoning attacks on FKGE.
- **Defense.** To mitigate the proposed attacks, we design a learning-based defense mechanism named LDM that seamlessly integrates with mainstream FKGE frameworks. The mechanism leverages both embedding density and anomaly detection models to accurately identify malicious clients.
- **Validation.** Theoretical analysis and empirical evidence show that untargeted poisoning attacks inflict substantial degradation on FKGE, with embedding-level poisoning particularly severe. Moreover, LDM effectively mitigates the proposed attacks and demonstrates strong generality and adaptability.

2 Related Work

2.1 Federated Knowledge Graph Embedding

FKGE aims to enhance the performance of local KGE models while preserving data privacy. FedE [5] adopts the strategy of averaging the embeddings of the same entities to achieve collaboration, while FedEC [6] leverages embedding-contrastive learning to handle data heterogeneity. FedLU [50] refines local embeddings through global embeddings and improves global embeddings via mutual knowledge distillation, thereby tackling multi-source data heterogeneity. FedEAN [23] tackles the zero-loss problem with entity-aware adversarial negative sampling, substantially enhancing the model's reasoning power. Although these works have effectively advanced FKGE research, the security issues therein have not been thoroughly explored, particularly regarding poisoning attacks.

2.2 Untargeted Poisoning Attacks and Defenses in Federated Learning

Untargeted poisoning attacks in FL compromise global models by manipulating model updates [21]. These attacks are categorized into data-level and model-level. While data poisoning attacks originate from supervised learning [36], model poisoning attacks emerge as a new security threat in FL. Without requiring operations on local training data, it demonstrates higher attack efficiency compared to data poisoning attacks [7, 27, 37]. Although poisoning in FKGE has been studied [47], research is limited to targeted attacks, overlooking untargeted poisoning attacks. FL defenses typically rely on robust aggregation [2, 11, 27, 40], which filters model updates using distance-based or statistical methods. However, these methods may fail or be difficult to apply directly in FKGE due to data heterogeneity across different clients, with inconsistent entity embeddings and incomplete entity overlaps. Therefore, designing the specific defense mechanism for FKGE is essential.

3 Problem Formulation

In this section, we provide the formulation of FKGE's system model and threat model of untargeted poisoning attacks.

3.1 System Model

Assume that there are K local clients in the system, and each client holds its own KG $\mathcal{G}_k = \{(h, r, t) | h, t \in E_k, r \in R_k\}$, where E_k and R_k denote the entity and relation sets. Each KG has a partially overlapping set of entities, but may have non-overlapping relationships. The entire dataset is denoted as $\mathcal{G} = \cup_{k=1,2,\dots,K} \mathcal{G}_k$. In FKGE, federated aggregation is performed at fixed intervals, and mainstream FKGE frameworks adopt ideas similar to FedAvg [22], such as FedE [5] and FedEC [6]. The aggregation can be defined as follows:

$$\mathbf{E}_k^n = \mathbf{P}_k \mathbf{E}^n, \quad \mathbf{E}^{n+1} = \left(1 \oslash \sum_{k=1}^K \mathbf{v}_k\right) \otimes \sum_{k=1}^K \mathbf{P}_k^T \mathbf{E}_k^{n+1}, \quad (1)$$

where $\mathbf{E}^n$ is the global embeddings in the n -th iteration, $\mathbf{P}_k$ is the mapping matrix from the global embeddings to local embeddings for client k and $\mathbf{P}_k^T$ is its transpose, $\mathbf{E}_k^n$ is the local embeddings for client k in the n -th iteration, $\mathbf{v}_k$ represents the entities existing in client k , $\mathbf{1}$ is a vector of all ones, $\oslash, \otimes$ represent element-wise division and multiplication respectively.

The detailed process is provided in Appendix A.

3.2 Threat Model

Attacker's goal. Attackers in FL could deliberately send incorrect model updates to sabotage the convergence and performance of the global model. Consequently, during FKGE training, attackers can influence the representational power of the global embeddings by sending incorrect local embeddings, thereby affecting other benign participants. The definition of untargeted poisoning attacks on FKGE is given in Definition 1.

DEFINITION 1 (UNTARGETED POISONING ATTACK ON FKGE). For a benign client, the entity embeddings uploaded to the server are normal, whereas a malicious client can upload arbitrary entity embeddings:

$$\text{Upload} = \begin{cases} *, & \text{if the } k\text{-th client is malicious} \\ \mathbf{E}_k, & \text{otherwise} \end{cases}. \quad (2)$$

where $*$ denotes the corrupted local entity embeddings.

Attacker's capability and knowledge Assume there are m malicious clients among K clients, where m is less than 50% of K ; otherwise, the global embeddings can be easily manipulated by the attackers. The attacker can access and poison the local KGs $\mathcal{G}_k$ of the compromised client, and it can also manipulate the entity embeddings $\mathbf{E}_k^{n+1}$ sent to the server. This is consistent with our considered two-level attacks (data level and embedding level). Note that we do not consider collusion among attackers, meaning that attackers do not collaborate to launch their attacks. For example, different attackers share the entity sets they possess. Attackers know the code, data, and embeddings of the clients they have manipulated. However, the attackers' knowledge is limited to the local training dataset and local embeddings of the client they are on. Moreover, since the attacks originate from the client side, we assume the server is benign, otherwise it would be impossible to achieve the objective of enhancing embedding representational power through FL using multi-source KGs.

4 Method

In this section, we elaborate on the proposed untargeted poisoning attacks against FKGE and introduce a learning-based defense mechanism named LDM to mitigate these attacks.

4.1 Framework Overview

Given that prior FKGE studies have neglected the crucial domain of untargeted poisoning attacks, we comprehensively design client-side poisoning attacks on FKGE at both **data and embedding levels**, encompassing one data poisoning attack (Sec. 4.2) and two embedding poisoning attacks (Sec. 4.3). The training of KGE models can be undermined by local poisoning methods and the aggregation of FL, ultimately affecting the application of downstream tasks. Subsequently, to counter these poisoning attacks, we introduce a **learning-based defense mechanism** that combines the density of entity embeddings with the anomaly detection model to identify and filter out malicious clients, thereby enhancing system robustness (Sec. 4.4). The overall framework is illustrated in Fig. 2 and more details will be explained as follows.

4.2 Data Poisoning Attack

FKGE dilutes the impact of malicious clients during the aggregation of different clients' entity embeddings, potentially rendering conventional poisoning attacks less effective. Consequently, we reconsider the characteristics of KG, aiming to generate poisoned data by substituting head and tail entities in triplets, which can be formulated as:

$$(h, r, t) \xrightarrow{\text{poison}} (h^*, r, t^*). \quad (3)$$

By replacing entities within original triples, we can simultaneously remove original knowledge while introducing new knowledge. To improve the efficacy of data-level attack and make it more interpretable, we propose a **powerful data poisoning attack** called *Similarity-based entity Replacement (SR)*. SR strategically selects the most dissimilar entities to replace the original entities in each triplet, aiming to maximize the disruption in the embedding space. The detailed process is as follows:

❶ **Acquisition of the entity embeddings.** For the k -th client, use clean local KGs to train the KGE model (such as TransE [3] or CompGCN [30]). Then, obtain the entity embeddings:

$$\mathbf{e}_i = \text{KGE}(e_i), \quad \forall e_i \in E_k. \quad (4)$$

where $\mathbf{e}_i$ is the embedding vector corresponding to entity e_i .

❷ **Entity similarity calculation.** Here, we use cosine similarity to measure the similarity between entity embeddings:

$$\cos \theta_{ij} = \frac{\mathbf{e}_i^T \mathbf{e}_j}{\|\mathbf{e}_i\| \|\mathbf{e}_j\|}, \quad \forall (e_i, e_j) \in E_k \times E_k. \quad (5)$$

❸ **Least similarity entity replacement.** Given the entity e_i that needs to be replaced, we need to find the entity that is least similar to it for replacement. Therefore, our goal is to minimize the similarity between the entity e_i and other entities:

$$e_i^* = \arg \min_{e_j \in E_k, e_j \neq e_i} \cos \theta_{ij}. \quad (6)$$

According to Eq. (6), we can obtain the poisoned triplet (h^*, r, t^*) by replacing the entities in the target triplet.

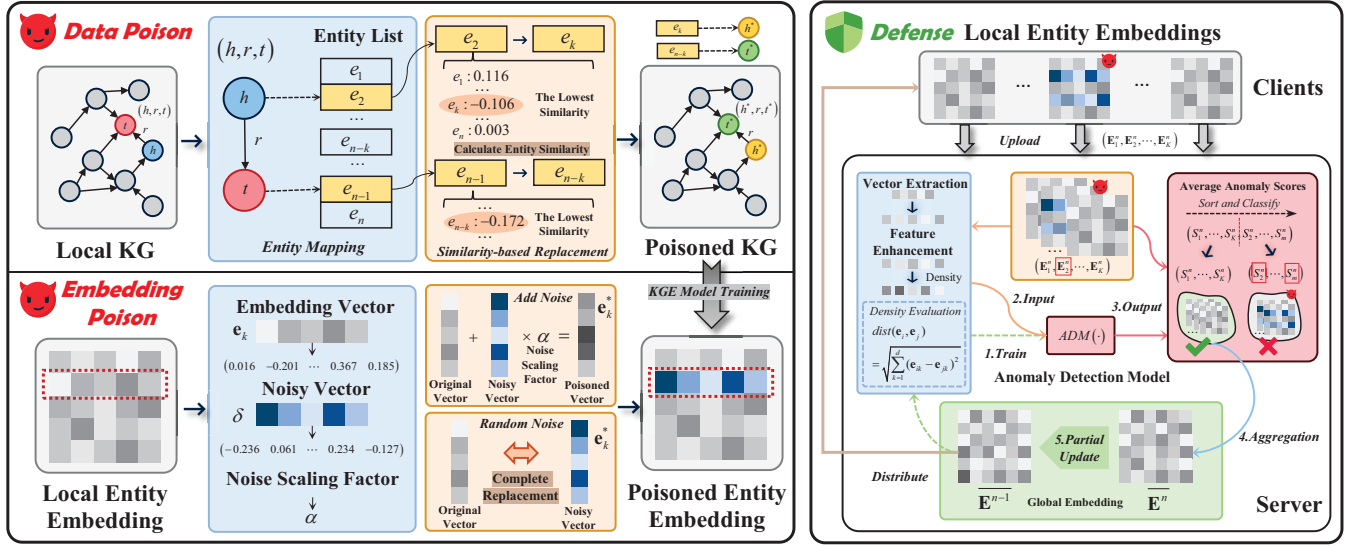


Figure 2: (Left) Illustration of the proposed untargeted poisoning attacks on FKGE. One data poisoning attack (refer to Sec. 4.2) and two embedding poisoning attacks (refer to Sec. 4.3). (Right) Overall process of learning-based defense mechanism against untargeted poisoning attacks (refer to Sec. 4.4).

4.3 Embedding Poisoning Attack

Previous attacks on KGE are limited to the data level, as tampering with local training data affects KGE model training. However, in FKGE, local entity embeddings serve as bridges for communication between clients. Therefore, we can directly attack the embeddings, which is more direct and effective than data poisoning attacks. The poisoning process can be defined as:

$$e_i \xrightarrow{\text{poison}} e_i^*, e_i \in E_k. \quad (7)$$

We design **two embedding poisoning attacks** against FKGE: *Add Noise to entity embeddings* (AN) and *replace entity embeddings with Random Noise* (RN):

AN introduces controlled perturbations by adding scaled random noise to the embedding vectors of a subset of entities. This noise is generated within the specified range $[-\beta, \beta]$ and scaled by a noise factor α , ensuring that only a portion of the embeddings are altered. Assuming the k -th client is a malicious client, the specific implementation steps of AN are as follows:

$$\delta_i \sim \mathcal{U}(-\beta, \beta)^d, \delta_i \in \mathbb{R}^d, \quad (8a)$$

$$e_i^* = e_i + \alpha \cdot \delta_i, \forall e_i \in E_p, \quad (8b)$$

where $p \in (0, 1]$ is the proportion of entities adding noise, $E_p \subset E_k$ is the set of selected entities, d is the dimension of the embedding vector corresponding to a single entity, $\mathcal{U}$ represents the uniform distribution, δ_i is the noise vector corresponding to the entity e_i , $\alpha > 0$ is the noise scaling factor, e_i and e_i^* are the normal and poisoned entity embedding vectors, respectively.

RN aggressively disrupts the embeddings by completely replacing the embedding vectors with randomly generated vectors within the specified range $[-\beta, \beta]$. This method aims to significantly disrupt the embedding space by making direct changes to all the entity embeddings. Compared to AN, RN can be defined as follows:

$$e_i^* = \delta_i, \forall e_i \in E_k. \quad (9)$$

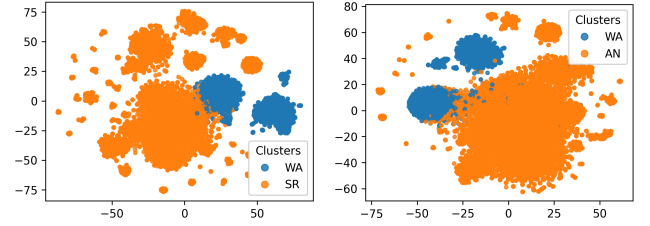


Figure 3: Defense Motivation. Comparison of t-SNE plots of the entity embeddings before (WA) and after (SR&AN) adversarial manipulation.

By directly tampering with the entity embeddings, it is possible to bypass the local training process without accessing the local data. For FKGE, such methods increase the efficiency of attacks and can significantly weaken the model's performance without being detected by local training.

4.4 Learning-based Defense Mechanism

Motivation. KGE achieves strong reasoning ability by capturing latent semantics through vector representations and using distance metrics and linear transformations for prediction. Untargeted poisoning attacks disrupt model performance by altering the embedding vector space at both data and embedding levels. As illustrated in Fig. 3, malicious manipulation increases the sparsity of entity embeddings. Motivated by this phenomenon, **embedding density** can serve as an indicator for detecting malicious clients [9].

We propose a multi-stage **Learning-based Defense Mechanism (LDM)**. First, vector extraction and density evaluation are performed on the embeddings. Next, feature representation is enhanced with density-based information. An anomaly detection model (e.g.,

Isolation Forest [20], ECOD [15]) trained on previous global embeddings analyzes current local embeddings, generating anomaly scores to identify and filter malicious clients [8]. The complete process follows these steps:

- 1 **Density evaluation.** We evaluate density by using the distance between the embedding vector and its neighboring vectors. For the KGE model E_k^n uploaded by the k -th client in the n -th communication round, we calculate the average Euclidean distance of the nearest λ vectors, and then use the overall average distance to represent the density of the embedding:

$$D_{ij} = \text{sort}(\text{dist}(\mathbf{e}_i, \mathbf{e}_j))[:\lambda] \\ \text{dist}(\mathbf{e}_i, \mathbf{e}_j) = \sqrt{\sum_{k=1}^d (\mathbf{e}_{ik} - \mathbf{e}_{jk})^2}, \mathbf{e}_j \in E_k^n, \mathbf{e}_j \neq \mathbf{e}_i \} \Rightarrow \bar{d}_i, \bar{d}_k, \quad (10)$$

where $\bar{d}_i = \frac{1}{\lambda} \sum_{j=1}^{\lambda} D_{ij}$, $\bar{d}_k = \frac{1}{N} \sum_{i=1}^N \bar{d}_i$.

- 2 **Embedding feature enhancement.** Based on the density evaluation results obtained previously, we enhance the features of the global embeddings E^{n-1} aggregated in the $(n-1)$ -th communication round and the local embeddings $\{E_k^n\}_{k=1}^K$ uploaded in the n -th communication round:

$$\hat{E}^{n-1} = E^{n-1} \cdot \frac{1}{K} \sum_{k=1}^K \bar{d}_k, \quad \hat{E}_k^n = E_k^n \cdot \bar{d}_k. \quad (11)$$

- 3 **Anomaly score calculation.** By using the enhanced global embedding $\hat{E}^n$ to train the anomaly model ADM, and then using the enhanced local embeddings $\{\hat{E}_k^n\}_{k=1}^K$ as input, we obtain the average anomaly scores for different clients:

$$\text{ADM}(\cdot) = \text{train}(\hat{E}^{n-1}), S_k = \text{ADM}(\hat{E}_k^n). \quad (12)$$

- 4 **Malicious client detection.** The mean of the anomaly scores is used as a threshold to identify malicious clients. Different anomaly detection models may have different characteristics; ECOD [15] is used as an example to illustrate the process:

$$\omega_k^n = \begin{cases} 0, & \text{if } S_k > \frac{1}{K} \sum_{k=1}^K S_k \\ 1, & \text{otherwise} \end{cases} \quad (13)$$

where ω_k^n represents the aggregation identifier of the client.

Clients are categorized into malicious and benign types through the above process. We employ a **retention mechanism** to obtain new global embeddings, where malicious clients inherit corresponding global embeddings from the previous round, while benign clients utilize newly uploaded embeddings:

$$E^n = \left(1 \odot \sum_{k=1}^K \mathbf{v}_k \right) \otimes \left(\underbrace{\sum_{k=1}^K \omega_k^n \mathbf{P}_k^T E_k^n}_{\text{benign}} + \underbrace{\sum_{k=1}^K (1 - \omega_k^n) \mathbf{P}_k^T E_k^{n-1}}_{\text{malicious}} \right). \quad (14)$$

The overall process is shown in Algorithm 1.

5 Theoretical Analysis

We provide theoretical justifications for the proposed attacks (SR, AN, RN) and the defense mechanism (LDM) from the following perspectives: (**Theorem 5.1 & Lemma 5.2**). Why are the proposed untargeted poisoning attacks effective? (**Theorem 5.3**). Why are

embedding-level attacks superior to data-level attacks? (**Theorem 5.4 & Theorem 5.5**). Why can the proposed defense scheme LDM effectively defend against such attacks?

THEOREM 5.1 (SR ATTACK EFFECTIVENESS). *Let the entity embedding matrix of clean client k be $E_k^{\text{clean}} \in \mathbb{R}^{|E_k| \times d}$. Under SR where a fraction $\pi_k \in (0, 1]$ of triples are poisoned by replacing head/tail entities with their least-similar counterparts, denote by E_k^{SR} the embeddings after local retraining. Assume the training operator $\mathcal{T}_k : \mathcal{G}_k \mapsto E_k$ is locally bi-Lipschitz w.r.t. the replacement discrepancy, and cosine dissimilarity is equivalent to Euclidean distance on a bounded (or normalized) domain. Then there exists $C_k > 0$ such that*

$$\mathbb{E}[\|E_k^{\text{SR}} - E_k^{\text{clean}}\|_F] \geq C_k \sqrt{\pi_k} \cdot \delta_k, \quad \delta_k \triangleq \min_{e_i \in E_k} \min_{e_j \neq e_i} \|\mathbf{e}_i - \mathbf{e}_j\|_2,$$

where the expectation is over the random selection of poisoned positions and minibatches.

The proof of Theorem 5.1 is deferred to Appendix C.2. It implies that SR yields a non-trivial local embedding shift scaling as $\sqrt{\pi_k}$ and lower-bounded by the client-wise separation δ_k (up to constants).

LEMMA 5.2 (NOISE-INDUCED PAIRWISE DISTANCE EXPANSION). *For any $\mathbf{e}_i, \mathbf{e}_j \in \mathbb{R}^d$ and $\delta_i, \delta_j \sim \mathcal{U}(-\beta, \beta)^d$, by AN/RN in Sec. 4.3, we have*

$$\mathbb{E}[\|\mathbf{e}_i^* - \mathbf{e}_j^*\|_2^2] = \begin{cases} \|\mathbf{e}_i - \mathbf{e}_j\|_2^2 + \frac{2\alpha^2 d \beta^2}{3}, & (\text{AN}), \\ \frac{2d\beta^2}{3}, & (\text{RN}). \end{cases}$$

(If AN perturbs only a ratio p of entities, the average increase over randomly selected entities/pairs scales with p up to constants.)

The proof of Lemma 5.2 is deferred to Appendix C.3. Lemma 5.2 characterizes the second-moment expansion injected by AN/RN, which is crucial for the aggregation error bound below.

THEOREM 5.3 (SUPERIORITY OF EMBEDDING-LEVEL ATTACKS). *Let $s \leq K$ be the number of clients that share a given entity in a round, and suppose $m \leq s$ of them are malicious. Each malicious client adopts AN or RN with independent noises as in Lemma 5.2. Let $\Delta = \mathbf{e}^{\text{glob}} - \mathbf{e}^{\text{clean}}$ be the aggregation error of this entity under Eq. (1) (mean aggregation on shared coordinates, so $a_k \asymp 1/s$). Then there exists $c_{\text{agg}} > 0$ such that*

$$\mathbb{E}\|\Delta\|_2^2 \gtrsim \left(\frac{m}{s^2}\right) c_{\text{agg}} \times \begin{cases} \frac{\alpha^2 d \beta^2}{3}, & (\text{AN}), \\ \frac{d \beta^2}{3}, & (\text{RN}). \end{cases}$$

(For AN with perturbation ratio p , the bound scales by a factor p for a randomly chosen entity.) Moreover, for any thresholds $\tau_{\text{emb}}, \tau_{\text{data}} > 0$ with $\tau_{\text{emb}}/\tau_{\text{data}}$ sufficiently large and any $\eta \in (0, 1)$, there exist $d_0(\eta)$ and $s_0(\eta)$ such that for all $d \geq d_0(\eta)$ and $s \geq s_0(\eta)$,

$$\mathbb{P}(\|\Delta\|_2 \geq \tau_{\text{emb}}) \geq 1 - \eta, \quad \mathbb{P}(\|\Delta_{\text{SR}}\|_2 \geq \tau_{\text{data}}) \leq \eta,$$

where Δ_{SR} denotes the corresponding aggregation error induced by SR on the same shared entity. In particular, under fixed attack ratios (e.g., m/s and π_k), embedding-level attacks exceed τ_{emb} with high probability when d (or noise intensity) is sufficiently large, while SR-induced error is suppressed as s grows due to averaging.

The proof of Theorem 5.3 is provided in Appendix C.4. Theorem 5.3 shows that AN/RN directly injects noise energy scaling with d and surviving aggregation via $\sum a_k^2 \asymp m/s^2$, whereas SR impacts uploads only indirectly through retraining and is further diluted by averaging.

THEOREM 5.4 (DENSITY SEPARATION). *Let $\bar{d}_k$ denote the average k NN distance in Eq. (10), and $\rho(\mathbf{E}_k^n) \triangleq 1/\bar{d}_k$ be a density proxy for client k at round n . Under (i) SR/AN/RN with non-degenerate parameters ($\alpha, \beta, \pi_k > 0$), (ii) mild regularity of benign embeddings (bounded domain and non-collapsing local neighborhoods) and (iii) assume the attack strength is non-negligible relative to the benign embedding scale (so that $\mu_{\text{malicious}} > \mu_{\text{benign}}$),*

$$\mathbb{E}[\rho(\mathbf{E}^{\text{malicious}})] < \mathbb{E}[\rho(\mathbf{E}^{\text{benign}})].$$

THEOREM 5.5 (LDM DEFENSE GUARANTEE UNDER SEPARATION). *Assume $m < K/2$ and the anomaly detector used by LDM is consistent on the density-enhanced features. Further assume a fixed mean-gap separation: there exists $\delta > 0$ such that, for sufficiently many entities and rounds,*

$$|\mathbb{E}[\rho(\mathbf{E}^{\text{benign}})] - \mathbb{E}[\rho(\mathbf{E}^{\text{malicious}})]| \geq \delta.$$

Then as the number of communication rounds increases, the probability that LDM correctly identifies malicious clients tends to 1.

The proofs of Theorem 5.4 and Theorem 5.5 are provided in Appendix C.5 and Appendix C.6, respectively. Overall, Theorem 5.4 establishes a stable density gap induced by attacks, and Theorem 5.5 shows that LDM exploits this gap with a consistent detector (and the $m < K/2$ regime) to drive misclassification probability to zero as rounds and samples grow.

6 Experiment

In this section, we verify the effectiveness of the overall framework. Specifically, we aim to address the following questions: **RQ1 (Attack Evaluation)**. Are the proposed data-level and embedding-level poisoning attacks effective? **RQ2 (Sensitivity Analysis)**. How do different experimental settings influence the effectiveness of the attacks, including the number of malicious clients (m), the intensity of the attacks (p & α), and the embedding dimension (d)? **RQ3 (Defense Validation)**. Can the proposed learning-based defense mechanism mitigate the attacks? **RQ4 (Defense Generalizability)**. Is the proposed defense mechanism general and adaptive? **RQ5 (Privacy Consideration)**. Can the proposed defense mechanism be combined with the privacy protection scheme?

6.1 Experiment Setup

Datasets and evaluation metrics. Traditional KG evaluation focuses on individual link prediction, but FKGE requires assessing link prediction across federated KG sets. In line with other FKGE literature [5, 6, 50], we use the FB15k237 dataset [29], divided into FB15k237-R5 and FB15k237-R10 for 5 and 10 clients, respectively (statistics in Appendix D.1). Link prediction is evaluated using Mean Reciprocal Rank (MRR) and Hits@N, where stronger attacks result in lower MRR and Hits@N. All results are reported after 200 iterations. See Appendix D for more details.

Baselines and implementation details. We apply the proposed attacks to FKGE algorithms: FedE [5], FedEC [6], and FedLU

[50], implemented on TransE [3] and RotatE [28]. We also integrate our defense mechanism into these baselines to assess its effectiveness. All configurations follow the FedLU framework [50] (see Appendix D.2). For embedding poisoning, the noise proportion p is 0.5, with $\alpha = 0.5$ and $\beta = 0.1$. LOF [4], Isolation Forest (IF) [20], ECOD[15] are used for anomaly detection. Experiments are performed with PyTorch [24] on NVIDIA RTX 4090 GPUs.

6.2 Attack Evaluation (RQ1&RQ2)

In this section, we demonstrate the effectiveness of the proposed attacks. First, we implement attacks on different baselines, datasets, and KGE models, and then test the link prediction performance on benign clients (**RQ1**). Finally, we configure different parameters to examine their impact on attack effectiveness. (**RQ2**).

Overall attack performance (RQ1). To validate attack effects, we set the malicious client ratio to 0.4 and conduct experiments across various datasets, baselines, and KGE models. Link prediction performance is evaluated using MRR, Hits@1 and Hits@10 under Non-attack (NA) and different attacks (SR, AN, RN). Results in Tab. 1 show that both data-level and embedding-level attacks significantly degrade performance, with embedding poisoning consistently outperforming data poisoning, supporting our analysis (Theorem 5.1, Lemma 5.2, Theorem 5.3) in Sec. 5. RotatE demonstrates greater robustness against data poisoning, with an average Hit@1 drop of only 12%, compared to 50% for TransE. This suggests that data poisoning attack effectiveness is model-dependent, influenced by relation modeling and loss function design. Conversely, embedding-level attacks are consistently effective across KGE models. Notably, while RotatE resists data poisoning, it is particularly vulnerable to embedding poisoning, likely due to its complex geometric modeling. Interestingly, AN outperforms RN for FedLU under TransE, potentially due to FedLU's resistance to local optimization drift, which diminishes the impact of random noise.

Comparison of different malicious clients (RQ2). We examine the impact of malicious client ratios on system performance using TransE and RotatE on FB15k237-R10. Experiments with 2-4 malicious clients show that performance degrades significantly as the ratio increases, highlighting the effectiveness of our attacks. Due to space limitations, the experimental results are deferred to Appendix D.3. For FedE and FedEC, we observe consistent behavioral patterns under different attacks. While the overall training process maintains steady progress, the system performance ultimately converges to suboptimal levels. Notably, under TransE, RN exhibits an interesting phenomenon of accelerating the early training process, suggesting a potential approach for improving FKGE's training efficiency. In contrast, FedLU displays a distinctive pattern: initial performance improvement followed by an abrupt decline at a certain stage. This behavior pattern validates that our proposed attacks achieve both stealthiness and effectiveness.

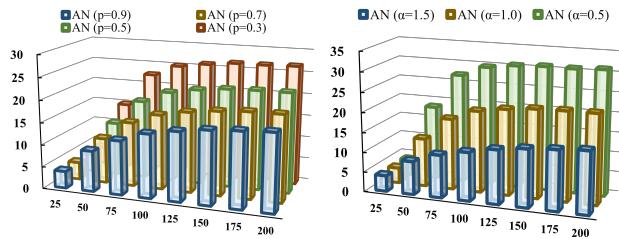
Comparison under different attack intensities (RQ2). We select AN to study the influence of different attack intensities. Specifically, the attack intensity of AN is determined by hyperparameters p and α . We set the malicious client ratio to 0.4 and use FB15k237-R10 and TransE to study system performance under different attack intensities ($p = 0.3, 0.5, 0.7, 0.9$ and $\alpha = 0.5, 1.0, 1.5$). The detailed results are shown in Fig. 4. It is intuitively evident that both an

Table 1: Comparison of link prediction performance under different attacks: Non-Attack (NA) and different attacks (SR, AN, RN), with malicious client ratio is 0.4. Bold and underline indicate best and second-best attack effects. The red values in the subscript show Hits@1 relative decrease (%). Please see details in Sec. 6.2.

Methods		NA			SR			AN			RN			
		Hits@1	Hits@10	MRR	Hits@1	Hits@10	MRR	Hits@1	Hits@10	MRR	Hits@1	Hits@10	MRR	
FB15k-237-R5	TransE	FedE	20.09	55.90	32.37	14.05 _{↓30%}	41.37	23.34	<u>11.57</u> _{↓42%}	<u>39.04</u>	<u>20.99</u>	7.14 _{↓64%}	23.05	12.68
		FedEC	21.37	52.28	31.86	16.91 _{↓21%}	45.36	26.68	<u>11.20</u> _{↓48%}	<u>37.86</u>	<u>20.33</u>	6.96 _{↓67%}	22.73	12.50
		FedLU	19.00	53.71	30.92	<u>2.45</u> _{↓87%}	36.48	14.16	1.35 _{↓93%}	<u>37.96</u>	<u>14.31</u>	11.79 _{↓38%}	49.11	24.82
	RotatE	FedE	24.93	61.34	37.37	21.82 _{↓12%}	52.14	32.15	<u>13.19</u> _{↓47%}	<u>42.17</u>	<u>23.23</u>	2.48 _{↓90%}	9.02	4.80
		FedEC	26.16	58.08	37.05	23.18 _{↓11%}	54.10	33.73	<u>12.14</u> _{↓54%}	<u>39.20</u>	<u>21.51</u>	2.43 _{↓91%}	8.70	4.70
		FedLU	22.57	55.48	33.71	21.05 _{↓7%}	52.25	32.90	<u>10.98</u> _{↓51%}	<u>50.00</u>	<u>24.89</u>	8.89 _{↓61%}	35.75	18.19
FB15k-237-R10	TransE	FedE	19.18	56.36	31.90	<u>11.43</u> _{↓40%}	45.83	23.52	13.00 _{↓32%}	<u>40.79</u>	<u>22.61</u>	10.13 _{↓47%}	27.93	16.17
		FedEC	20.74	56.68	33.12	14.43 _{↓30%}	50.64	27.12	<u>12.87</u> _{↓38%}	<u>39.90</u>	<u>22.18</u>	10.07 _{↓51%}	28.04	16.11
		FedLU	17.34	53.15	29.81	1.74 _{↓90%}	<u>41.37</u>	<u>15.57</u>	<u>2.62</u> _{↓85%}	34.96	14.13	8.99 _{↓48%}	43.91	20.78
	RotatE	FedE	23.19	60.91	36.24	18.67 _{↓19%}	53.34	30.66	<u>13.68</u> _{↓41%}	<u>40.40</u>	<u>22.88</u>	3.26 _{↓86%}	12.35	6.38
		FedEC	22.47	62.19	36.22	18.86 _{↓16%}	57.31	32.24	<u>12.76</u> _{↓43%}	<u>37.86</u>	<u>21.37</u>	3.20 _{↓86%}	12.18	6.30
		FedLU	21.86	55.03	33.11	18.73 _{↓14%}	51.36	29.52	<u>11.92</u> _{↓45%}	<u>48.52</u>	<u>24.47</u>	7.55 _{↓65%}	30.37	15.24

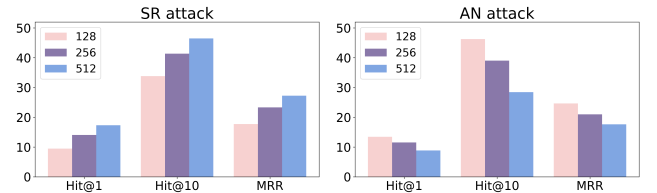
Table 2: Ablation study on FB15k-237-R10 with/without LDM. Bold indicates the best performance under the current baseline algorithm and KGE model. The blue values in the subscript indicate the relative increase in the Hits@1 metric.

Methods		NA			SR			AN			RN		
		Hits@1	Hits@10	MRR	Hits@1	Hits@10	MRR	Hits@1	Hits@10	MRR	Hits@1	Hits@10	MRR
TransE	FedE	19.18	56.36	31.90	11.43	45.83	23.52	13.00	40.79	22.61	10.13	27.93	16.17
	FedE+LDM	–	–	–	16.57 _{↑27%}	54.47	29.71	19.82 _{↑36%}	57.50	33.00	19.87 _{↑51%}	57.62	33.00
	FedEC	20.74	56.68	33.12	14.43	50.64	27.12	12.87	39.90	22.18	10.07	28.04	16.11
	FedEC+LDM	–	–	–	17.27 _{↑14%}	57.03	31.10	20.18 _{↑35%}	57.90	33.32	20.22 _{↑49%}	57.98	33.36
	FedLU	17.34	53.15	29.81	1.74	41.37	15.57	2.62	34.96	14.13	8.99	43.91	20.78
	FedLU+LDM	–	–	–	14.93 _{↑76%}	53.09	28.36	18.22 _{↑90%}	55.00	31.12	17.90 _{↑51%}	55.19	31.02
RotatE	FedE	23.19	60.91	36.24	18.67	53.34	30.66	13.68	40.40	22.88	3.26	12.35	6.38
	FedE+LDM	–	–	–	21.01 _{↑10%}	58.20	33.89	24.47 _{↑47%}	59.51	36.54	24.66 _{↑92%}	59.38	36.62
	FedEC	22.47	62.19	36.22	18.86	57.31	32.24	12.76	37.86	21.37	3.20	12.18	6.30
	FedEC+LDM	–	–	–	20.14 _{↑6%}	58.10	33.32	24.89 _{↑54%}	59.73	36.89	24.89 _{↑97%}	59.63	36.90
	FedLU	21.86	55.03	33.11	18.73	51.36	29.52	11.92	48.52	24.47	7.55	30.37	15.24
	FedLU+LDM	–	–	–	20.46 _{↑8%}	52.08	31.02	23.66 _{↑54%}	56.37	34.79	23.25 _{↑72%}	57.62	34.88

**Figure 4: Comparison of attack effects under different attack intensities. (Left) Varying the proportion p of entities perturbed with noise. (Right) Varying the noise scaling factor α . The x-axis is epochs and the y-axis is MRR.**

increase in the proportion of entities adding noise and an amplification of the noise scaling factor lead to enhanced attack strength, resulting in more severe degradation of system performance.

Ablation study on embedding dimension (RQ2). We vary only the embedding dimension $d \in \{128, 256, 512\}$ on FB15k-237-R5 and TransE ($m/K = 0.4$) and evaluate SR (data-level) and AN

**Figure 5: Ablation on embedding dimension, highlighting that data-level and embedding-level attacks exhibit different trends as the dimension varies. See details in Sec. 6.2.**

(embedding-level). As shown in Fig. 5, we observe a consistent pattern: larger d mitigates SR but exacerbates AN. A plausible explanation is that higher dimensionality provides representational slack that buffers SR’s mismatch between structure and semantics, whereas under AN the noise accumulates in ℓ_2 norm approximately as $\sqrt{d}$ at fixed magnitude, reducing the signal to noise ratio and making higher dimensional embeddings more brittle.

6.3 Defense Evaluation (RQ3&RQ4)

In this section, we verify whether the proposed defense mechanism is effective against the proposed attacks. First, we investigate

the performance of different baseline algorithms when encountering attacks after incorporating the proposed defense mechanism (RQ3). Then, we replace the anomaly detection model to verify the adaptability of our proposed defense framework (RQ4).

Overall defense effectiveness (RQ3). Following the attack settings in RQ1, we evaluate baseline algorithms on FB15k237-R10 against SR, AN, and RN attacks, before and after integrating LDM. The ablation study (Tab. 2) shows that LDM significantly enhances performance across all baselines, which proves its general effectiveness and is consistent with our analysis in Sec. 5 (Theorem 5.4, Theorem 5.5). For data poisoning attack (SR), the final performance we achieve is on average about 11.7% lower than the optimal performance. This is due to data poisoning corrupting normal data, leading to performance degradation from lost information utilization, not incorrect filtration of malicious clients. For embedding poisoning attacks (AN, RN), we are able to achieve better performance than original performance through this mechanism, with an average improvement of approximately 4.8%. This may be due to the following two reasons: I) The noise injected by the attacks can indirectly improve the link prediction performance; II) Our proposed defense mechanism can filter out poorly performing benign clients while filtering out malicious clients.

Adaptability of defense framework (RQ4). To further verify the superiority of the proposed mechanism, we replace the adopted anomaly detection models to demonstrate the adaptability of our defense mechanism. By successively employing three anomaly detection models (LOF [4], Isolation Forest (IF) [20], ECOD[15]), we verify the defense effectiveness of the FedE with TransE against untargeted poisoning attacks on FB15k237-R10. As shown in Tab. 3, regardless of which anomaly detection model is employed, LDM achieves similar effects in defending against the proposed attacks and effectively enhancing system robustness.

6.4 Considerations for Privacy (RQ5)

While FKGE effectively facilitates collaborative learning through entity embeddings transmission, this mechanism raises privacy concerns [12, 31, 32, 42]. To address this, integrating privacy-preserving mechanism is essential. Our approach is highly adaptable and can seamlessly work alongside various privacy-preserving mechanisms. To validate this compatibility, we conduct experiments under three conditions: **I)** Vanilla: FGKE combined with LDM, without any privacy protection methods. **II)** Noise: Add random noise $[-0.01, 0.01]$ to entity embeddings. **III)** DPSGD [1]: Differentially private stochastic gradient descent with $(\epsilon=32, \delta=1e-5)$ in local training. The results in Tab. 4 demonstrate that LDM maintains its effectiveness with privacy-preserving mechanisms. This adaptability highlights LDM's potential to simultaneously address both **security** and **privacy** in a unified framework. We discuss the impact of privacy protection mechanisms on the **detection accuracy** (Precision, Recall, F1 Score) of LDM in Appendix D.4.

6.5 Handling False Positives

Our **retention mechanism** in Eq. (14) serves dual purposes: protecting against malicious updates while mitigating the impact of false positives. If a benign client is falsely flagged as malicious due to inherently diverse or unstable training, its prior embeddings are preserved rather than discarded, allowing potential recovery

Table 3: Evaluation of defense effectiveness using different anomaly detection models on FB15k-237-R10.

Attacks	ADM	Hit@1	Hit@10	MRR
FedE+TransE	SR	LOF	15.86 $\uparrow 23\%$	53.59
		IF	16.56 $\uparrow 27\%$	54.66
		ECOD	16.57 $\uparrow 27\%$	54.47
	AN	LOF	20.19 $\uparrow 37\%$	57.63
		IF	19.94 $\uparrow 36\%$	57.57
		ECOD	19.82 $\uparrow 36\%$	57.50
	RN	LOF	19.85 $\uparrow 51\%$	57.56
		IF	20.27 $\uparrow 53\%$	57.52
		ECOD	19.87 $\uparrow 51\%$	57.62

Table 4: Evaluation of defense effectiveness with different privacy-preserving schemes on FB15k-237-R10.

Attacks	Privacy	Hit@1	Hit@10	MRR
FedE+TransE	SR	Vanilla	16.57 $\uparrow 27\%$	54.47
		Noise	16.16 $\uparrow 25\%$	52.88
		DPSGD	15.12 $\uparrow 18\%$	50.52
	AN	Vanilla	19.82 $\uparrow 36\%$	57.50
		Noise	19.51 $\uparrow 34\%$	56.04
		DPSGD	19.84 $\uparrow 36\%$	57.18

Table 5: Ablation on the retention mechanism (RM) in Eq. (14), highlighting its effectiveness in mitigating false positives. See details in Sec. 6.5.

Attacks	Hit@1	Hit@10	MRR
FedE+TransE	SR (wo. RM)	15.39 $\uparrow 21\%$	53.01
	SR (w. RM)	16.57 $\uparrow 27\%$	54.47
	AN (wo. RM)	19.79 $\uparrow 35\%$	57.37
	AN (w. RM)	19.82 $\uparrow 36\%$	57.50

subsequently. We conduct an **ablation study** on FB15k-237-R10 to demonstrate its effectiveness, as shown in Tab. 5. Incorporating this mechanism makes the defense more effective (especially for SR), but the original framework also demonstrates excellent performance.

7 Conclusion

In this paper, we comprehensively investigate untargeted poisoning attacks on FKGE from both attack and defense perspectives, addressing an unexplored area in FKGE research. Regarding attacks, we rethink the weaknesses of FKGE and propose attacks at both data (SR) and embedding levels (AN, RN). In terms of defense, we propose LDM to mitigate the proposed untargeted poisoning attacks, which combines the embedding density and anomaly detection model. Furthermore, our theoretical analysis establishes the effectiveness of the proposed attacks and defense, especially demonstrating the superiority of embedding-level attacks. Experimental results align with the theoretical analysis, corroborating the destructive impact of the proposed attacks on FKGE and the effectiveness of LDM in defending against untargeted poisoning.

Acknowledgments

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A Process of Federated Knowledge Graph Embedding

The detailed process is illustrated as follows:

- **Step I.** In the n -th iteration, the server sends the corresponding portions of the global embeddings $\mathbf{E}^n$ to the local clients based on their respective mappings $\{\mathbf{P}_k\}_{k=1}^K$.
- **Step II.** Each client updates its local relation and entity embeddings $\mathbf{R}_k^n$ and $\mathbf{E}_k^n$ based on its own KG $\mathcal{G}_k$, yielding updated embeddings $\mathbf{R}_k^{n+1}$ and $\mathbf{E}_k^{n+1}$. Then, the clients upload their entity embeddings to the server.
- **Step III.** The server maps the local embeddings $\{\mathbf{E}_k^{n+1}\}_{k=1}^K$ uploaded by clients to their corresponding positions in the global embeddings, and then obtains the new global embedding $\mathbf{E}^{n+1}$ based on the specified aggregation rule.

The fundamental distinction between FKGE and traditional FL lies in FKGE’s need to handle non-IID graph-structured data, while also achieving partial entity alignment across distributed clients through dynamic mapping mechanisms. As a result, conventional parameter aggregation techniques commonly used in FL are inadequate for FKGE, as highlighted in prior studies [5, 6, 50].

B Algorithm Pseudocode

This paper focuses on unveiling the vulnerability of FKGE to untargeted poisoning attacks and designing corresponding defense mechanism to mitigate such attacks. Thus, the overall framework comprises two integrated components: the attack module (to systematically inject perturbations into local KG embeddings) and the defense module (to detect and eliminate adversarial embeddings).

C Supplementary Theoretical Analysis

C.1 Standing Assumptions and Notation.

We assume the local training operator $\mathcal{T}_k : \mathcal{G}_k \mapsto E^{(k)}$ (optimizer & schedule fixed) satisfies a *bi-Lipschitz calmness* property with respect to a graph discrepancy d_G that counts head/tail replacements with appropriate weights: there exist $0 < \underline{L}_k \leq \bar{L}_k < \infty$ such that for any $\mathcal{G}, \mathcal{G}'$ reachable by finitely many entity replacements,

$$\underline{L}_k d_G(\mathcal{G}, \mathcal{G}') \leq \|\mathcal{T}_k(\mathcal{G}) - \mathcal{T}_k(\mathcal{G}')\|_F \leq \bar{L}_k d_G(\mathcal{G}, \mathcal{G}'). \quad (15)$$

This holds locally under PL-type dynamics with bounded gradients and a fixed number of steps. For k NN density (Eq. (10)), we use its

bounded-difference property: changing one point affects at most $O(k/N)$ terms in the sample mean of k NN distances, enabling McDiarmid/Bernstein concentration.

C.2 Proof details for Theorem 5.1

PROOF. By SR, an entity e_i in a poisoned triple is replaced by its least-similar counterpart in cosine (Eq. (6)), which on a bounded (or normalized) domain is equivalent to inducing a large Euclidean separation. Define the client-wise *minimal nearest-neighbor separation* as

$$\delta_k \triangleq \min_{e_i \in E_k} \min_{j \neq i} \|e_i - e_j\|_2.$$

For any replaced entity e_i , since SR chooses a least-similar counterpart, the realized replacement distance satisfies

$$\|e_i - e_i^*\|_2 \geq \min_{j \neq i} \|e_i - e_j\|_2 \geq \delta_k.$$

When a fraction π_k of triples are SR-poisoned, let M be the (random) number of replacement events (heads/tails) induced by these poisoned triples. Under the discrepancy definition d_G (a weighted ℓ_2 -type aggregation of replacement magnitudes), Cauchy–Schwarz over the M replacement events yields

$$d_G(\mathcal{G}_k^{\text{SR}}, \mathcal{G}_k^{\text{clean}}) \geq c_0 \sqrt{M} \delta_k \geq c'_0 \sqrt{\pi_k} \delta_k,$$

for some constants $c_0, c'_0 > 0$ where c'_0 absorbs the deterministic scaling between M and π_k (e.g., dataset size and head/tail weighting in d_G). Applying the lower inequality in Eq. (15),

$$\|E_k^{\text{SR}} - E_k^{\text{clean}}\|_F \geq \underline{L}_k d_G(\mathcal{G}_k^{\text{SR}}, \mathcal{G}_k^{\text{clean}}) \geq (\underline{L}_k c'_0) \sqrt{\pi_k} \delta_k.$$

Taking expectation over the random choice of poisoned positions preserves the bound, giving the stated $C_k = \underline{L}_k c'_0$ rate and the $\sqrt{\pi_k}$ scaling. $\square$

Algorithm 1: Untargeted poisoning attacks on FKGE and corresponding defense mechanism

Input: Training Rounds N , K clients with local KGs $\{\mathcal{G}_k\}_{k=1}^K$ (with m malicious clients included)

- 1 Server initializes $\mathbf{E}^0$ and broadcasts $\{\mathbf{E}_k^0\}_{k=1}^K$ to clients
- 2 **for** $n = 1, \dots, N$ **do**

// Attacks (Section 4.2 and 4.3)

- 3 **for** Client $k \in \{1, 2, \dots, K\}$ **do**
- 4 **if** k -th client is malicious (SR) **then**
- 5 $(h, r, t) \xrightarrow{\text{poison}} (h^*, r, t^*)$ in Eq. (4)-Eq. (6)
- 6 KGE model training using $\mathcal{G}_k$
- 7 **if** k -th client is malicious (AN & RN) **then**
- 8 $e_i \xrightarrow{\text{poison}} e_i^*, e_i \in E_k$ in Eq. (8)-Eq. (9)
- 9 Upload $\mathbf{E}_k^n$ to Server

// Defense (Section 4.4)

- 10 **for** Client $k \in \{1, 2, \dots, K\}$ **do**
 - 11 $\bar{d}_k \leftarrow (\mathbf{e}_k^n, N, \lambda)$ in Eq. (10)
 - 12 $S_k \leftarrow (\mathbf{E}^{n-1}, \mathbf{E}_k^n, \bar{d}_k, K)$ in Eq. (11) - Eq. (12)
 - 13 $\omega_k^n \leftarrow (S_k, K)$ in Eq. (13)
 - 14 $\mathbf{E}^n \leftarrow (\mathbf{E}^{n-1}, \mathbf{E}_k^n, \omega_k^n, \mathbf{P}_k, \mathbf{v}_k, K)$ in Eq. (14)
-

C.3 Proof details for Lemma 5.2

PROOF. Let $\delta_i, \delta_j \stackrel{\text{i.i.d.}}{\sim} \mathcal{U}(-\beta, \beta)^d$ with $\mathbb{E}[\delta] = \mathbf{0}$ and $\text{Var}[\delta_i] = \beta^2/3$. For AN ($\mathbf{e}_u^* = \mathbf{e}_u + \alpha\delta_u$),

$$\|\mathbf{e}_i^* - \mathbf{e}_j^*\|_2^2 = \|\mathbf{e}_i - \mathbf{e}_j\|_2^2 + 2\alpha(\mathbf{e}_i - \mathbf{e}_j)^\top (\delta_i - \delta_j) + \alpha^2 \|\delta_i - \delta_j\|_2^2,$$

whose expectation equals $\|\mathbf{e}_i - \mathbf{e}_j\|_2^2 + \frac{2\alpha^2 d \beta^2}{3}$ since the cross term vanishes and $\mathbb{E}\|\delta_i - \delta_j\|_2^2 = 2d\beta^2/3$. For RN ($\mathbf{e}_u^* = \delta_u$), independence yields $\mathbb{E}\|\mathbf{e}_i^* - \mathbf{e}_j^*\|_2^2 = 2d\beta^2/3$.

Remark (partial AN with ratio p). If AN perturbs only a proportion p of entities (as in Eq. (8)), then the above expansion holds conditionally on the involved entities being perturbed; when the pair (i, j) is drawn uniformly at random, the additive increase scales as $\Theta(p\alpha^2 d \beta^2)$ up to constants. $\square$

C.4 Proof details for Theorem 5.3

PROOF. Let an entity be (partially) shared by the client set $\mathcal{S}$, $|\mathcal{S}| = s$. The global update for that coordinate is linear in local contributions (Eq. (1)):

$$\mathbf{e}_{n+1}^{\text{glob}} - \mathbf{e}_{\text{clean}, n+1}^{\text{glob}} = \Delta = \sum_{k \in \mathcal{S}} a_k P_k^\top \xi_k, \quad \sum_{k \in \mathcal{S}} a_k = 1,$$

where ξ_k is the attack perturbation ($\mathbf{0}$ for benign) and P_k selects the entity coordinate(s). Because P_k is a selection/reindexing map, its smallest singular value on the selected coordinates is 1, so $\|P_k^\top x\|_2 = \|x\|_2$ for those coordinates.

With m malicious clients:

$$\mathbb{E}\|\Delta\|_2^2 = \sum_{k \text{ mal}} a_k^2 \mathbb{E}\|\xi_k\|_2^2 \geq \left(\min_{k \text{ mal}} \mathbb{E}\|\xi_k\|_2^2 \right) \sum_{k \text{ mal}} a_k^2.$$

Under mean aggregation on shared entities, $a_k \asymp 1/s$, hence $\sum_{k \text{ mal}} a_k^2 \asymp m/s^2$.

For AN, $\xi_k = \alpha\delta_k$ with $\delta_k \sim \mathcal{U}(-\beta, \beta)^d$, so $\mathbb{E}\|\xi_k\|_2^2 = \alpha^2 \mathbb{E}\|\delta_k\|_2^2 = \alpha^2 d \beta^2/3$. For RN, the uploaded embedding is replaced by δ_k ; relative to the clean upload, one can write $\xi_k = (\delta_k - \mathbf{e}_k^{\text{clean}})$. Conditioned on $\mathbf{e}_k^{\text{clean}}$, δ_k is independent and zero-mean, so

$$\mathbb{E}\|\xi_k\|_2^2 = \mathbb{E}\|\delta_k - \mathbf{e}_k^{\text{clean}}\|_2^2 = \mathbb{E}\|\delta_k\|_2^2 + \|\mathbf{e}_k^{\text{clean}}\|_2^2 \geq d\beta^2/3.$$

Therefore, for some $c_{\text{agg}} > 0$ (absorbing constants from aggregation bookkeeping),

$$\mathbb{E}\|\Delta\|_2^2 \gtrsim \left(\frac{m}{s^2} \right) c_{\text{agg}} \times \begin{cases} \frac{\alpha^2 d \beta^2}{3}, & \text{AN,} \\ \frac{d \beta^2}{3}, & \text{RN.} \end{cases}$$

(For AN with perturbation ratio p , the bound scales by a factor p for a randomly chosen entity.)

High-probability comparison. Since Δ is a weighted sum of independent bounded noises, $\|\Delta\|_2^2$ concentrates around its mean with variance growing at most linearly in d . Thus, for any fixed $\tau_{\text{emb}} > 0$ and $\eta \in (0, 1)$, there exists $d_0(\eta)$ (and $s_0(\eta)$ for the weight regularity) such that for all $d \geq d_0(\eta)$ and $s \geq s_0(\eta)$,

$$\mathbb{P}(\|\Delta\|_2 \geq \tau_{\text{emb}}) \geq 1 - \eta.$$

In contrast, SR affects embeddings indirectly via retraining; Theorem 5.1 gives a per-client deviation $\Omega(\sqrt{\pi_k} \delta_k)$. When aggregating a shared entity over s clients by averaging, the induced coordinate-level error Δ_{SR} is further attenuated by a factor $\asymp 1/s$, hence for

any fixed $\tau_{\text{data}} > 0$,

$$\mathbb{P}(\|\Delta_{\text{SR}}\|_2 \geq \tau_{\text{data}}) \leq \eta \quad \text{for all sufficiently large } s.$$

This proves that embedding-level attacks can be strictly more damaging than SR in typical FKGE regimes (moderate/large d with partial sharing). $\square$

C.5 Proof details for Theorem 5.4

PROOF. Let $\bar{d}_k$ be the client-wise mean of k NN distances (Eq. (10)), and $\rho(\mathbf{E}^{(k)}) \triangleq 1/\bar{d}_k$ the density proxy. Under AN/RN, Lemma 5.2 implies an expansion of pairwise distances in expectation; SR pushes entities toward far neighbors by construction (Theorem 5.1). Assume the attack strength is non-negligible relative to benign embedding scale so that the population mean k NN distance satisfies $\mu_{\text{malicious}} > \mu_{\text{benign}}$ (for the considered non-degenerate parameters). The k NN mean has bounded differences ($O(k/N)$), hence by McDiarmid/Bernstein,

$$\mathbb{P}(|\bar{d}_k - \mu_k| \leq \varepsilon) \rightarrow 1 \quad (N \rightarrow \infty)$$

for any fixed $\varepsilon \in (0, \frac{1}{2}(\mu_{\text{malicious}} - \mu_{\text{benign}}))$. Since $x \mapsto 1/x$ is decreasing on $(0, \infty)$, with high probability $\rho(\mathbf{E}^{\text{malicious}}) < \rho(\mathbf{E}^{\text{benign}})$, hence also $\mathbb{E}[\rho(\mathbf{E}^{\text{malicious}})] < \mathbb{E}[\rho(\mathbf{E}^{\text{benign}})]$. $\square$

C.6 Proof details for Theorem 5.5

PROOF. LDM first forms density-enhanced embeddings (Eq. (11)), trains a (fixed-choice) anomaly detector ADM on the previous round's enhanced global embedding and evaluates current clients to obtain scores S_k (Eq. (12)), then identifies malicious clients by a thresholding rule (e.g., the mean-based ECOD criterion in Eq. (13)), and finally aggregates with a retention mechanism (Eq. (14)), which reuses the previous global coordinates for flagged clients while admitting only benign uploads.

Assume $m < K/2$, ADM is risk-consistent on the density-enhanced features, and the density scores satisfy a fixed mean-gap separation $\delta > 0$ as in Theorem 5.5. Under this separation and concentration over growing (entities $\times$ rounds) samples, the score distributions become increasingly separable, so the induced Bayes error tends to 0. Risk-consistency then implies the empirical false positive/negative rates both converge to 0. Because Eq. (14) prevents error accumulation—flagged (suspected-malicious) coordinates revert to the previous global state—the effect of sporadic misclassifications is confined to the round they occur. Therefore, the probability that LDM correctly identifies malicious clients tends to 1 as rounds and entity counts grow. $\square$

D Experimental Supplement

D.1 Dataset Statistics

In this paper, we utilize the widely adopted FB15k237 knowledge graph (KG) dataset [29] as the foundational dataset, which is partitioned into two federated subsets—FB15k237-R5 and FB15k237-R10—corresponding to 5 and 10 clients, respectively. The detailed statistics of these two subsets are presented in Table 6. This partitioning strategy reflects the level of data heterogeneity commonly addressed in FKGE literature [5, 6], aligning our experimental setup with standard practices and reinforcing the robustness and relevance of our research framework.

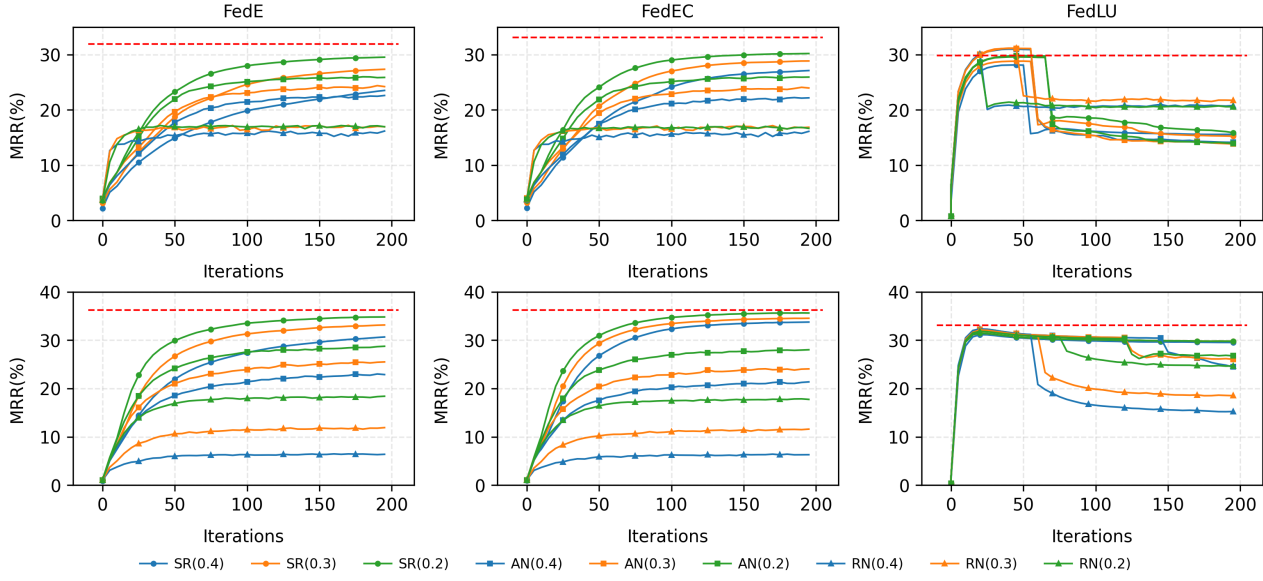


Figure 6: Comparison of attack effects under different ratios (0.4/0.3/0.2) of malicious clients. The top results are for TransE, while the bottom results are for RotatE. The red dotted line represents the system’s original performance without attacks.

Table 7: Metrics of anomaly detection model with different privacy-preserving schemes on FB15k-237-R10.

	Attacks	Precision	Recall	F1 Score
FedE+LDM	SR	77.65%	95.60%	85.68%
	SR+Noise	77.38%	96.63%	85.93%
	SR+DPSGD	50.59%	69.13%	58.40%
	AN	100.00%	100.00%	100.00%
	AN+Noise	100.00%	100.00%	100.00%
	AN+DPSGD	100.00%	100.00%	100.00%

Table 6: Statistics of FB15k-237-R5 and FB15k-237-R10.

Datasets	Clients	Relations	Entities	Triplets
FB15k-237-R5	1	47	10756	46549
	2	47	10554	53115
	3	47	12692	64375
	4	47	11091	73652
	5	49	12273	72425
FB15k-237-R10	1	23	6969	17940
	2	23	7237	19860
	3	23	8897	36468
	4	23	7947	24895
	5	23	10072	54895
	6	23	7981	31902
	7	23	10435	30654
	8	23	8231	28298
	9	23	9874	22477
	10	30	8555	42727

D.2 Hyperparameter Setting

We employ the Adam [13] optimizer with a learning rate of $1e-4$. For all methods, the local model is trained for 3 epochs during each communication round, with validation set evaluations occurring every 5 rounds. The client sampling fraction in each communication

round for FKGE learning is set to 100%. For the KGE models, we configure the batch size to 1,024, the negative sampling size to 256, and set the dimensionality of entity and relation embeddings to 256. Regarding specific method configurations: For FedEC, the hyperparameters are configured with $\tau = 0.2$ and $\mu_{con} = 0.3$. For FedLU, the distillation coefficient $\mu_{con} = 2$.

Regarding the attack settings, for SR, we perform entity replacement on half of the data, and the IDs of the malicious agents are set as 1, 4 (for 5 clients), and 1, 4, 7, 10 (for 10 clients). For AN and RN, we set to add noise or replace with noise for half of the embeddings. For AN, the settings are $\alpha = 0.5$ and $\beta = 0.1$; for RN, the setting is $\beta = 0.1$. The IDs of the malicious agents are set as 1, 4 (for 5 clients), and 1, 5, 6, 10 (for 10 clients).

D.3 Comparison of Different Malicious Clients

We investigate the impact of different ratios of malicious clients on attack effectiveness, as shown in Fig. 6. When the ratio of malicious clients increases from 0.2 to 0.4, the attack effectiveness notably improves, which aligns with intuition since a larger number of attackers implies more potent attack capabilities.

D.4 The Accuracy Rate of LDM

Here, we analyze the average metrics (Precision, Recall, F1 score) of the anomaly detection model across all detection iterations (Tab. 7). For AN, the addition of the privacy protection mechanism has no impact. However, for SR, since DPSGD introduces differential privacy during the local training process, this will obscure the uploaded entity embeddings, thereby causing a decline in the accuracy of LDM recognition. However, due to the excellence of the overall framework (the existence of the retention mechanism (Tab. 5), LDM can still effectively reduce the effectiveness of the attack (Tab. 4). Future research can delve deeper into the trade-off between privacy budget and defense effectiveness.